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# FIOD-VUE: Focusing on Invariant Information in Object Detection of Varying Underwater Environment

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***Abstract*—The varying environmental conditions pose challenges to existing object detection methods as they lead to changes in the overall feature distribution of images. Underwater images are particularly susceptible to environmental conditions changes, resulting in phenomena like color deviation. This paper propose an object detection model, FIOD-VUE, which focuses on invariant information across different underwater environments to enhance the model's generalization capability. Inspired by frequency domain analysing, we design a Frequency-Invariant Attention (FIA) module. This module use frequency filters to focus on specific frequency signals, i.e., cross-domain invariant information. Additionally, we design the Multi-scale Image-level Feature Alignment (MIFA) to adaptively adjust the frequency filters in the FIA and assist the backbone in extracting domain-confusion features. Through adversarial training, the distribution gap between the source domain and target domain is reduced. To enrich the domain shift database, we also afford an HD-Deepfish dataset. Numerous experiments on the S-UODAC2020 and the HD-Deepfish datasets were executed and yielded impressive results, with average precision (AP) scores of 56.8% and 37.1%, respectively, surpassing the performance of the existing underwater object detection (UOD) models. The link of the code is released at: https://github.com/JOU-UIP/FIOD-VUE.**

***Index Terms*—Underwater, Object detection, Domain adaptation, Frequency**

## I. Introduction

DEEP learning-based object detection has made significant advances, becoming an indispensable component in numerous real-world applications such as video surveillance, augmented reality, autonomous navigation, human-computer interaction, and self-checkout convenience stores [1]. Most state-of-the-art object detection models relyed on large amounts of annotated data to improve performance [2]–[4]. However, these models typically assumed that the training source data and target data are independent and identically distributed (i.i.d.) [5]. In reality, changes in environmental conditions often lead to inconsistencies between the source and target domain distributions, as shown in Fig. 1. In the HD-Deepfish dataset, there are significant differences in the gaussian distribution of color hues between the target domain's style3 and the source domain's style1 and style2. When applying a pre-trained detector in the target domain, significant performance degradation is inevitable. Particularly in underwater environments, influenced by water absorption and scattering, the captured underwater images exhibit different color biases. This phenomenon is known as domain shift or distribution shift, which is quite common in underwater deployment scenarios. The quantitative performance comparison results on the HD-Deepfish dataset are illustrated in Fig. 2. Although some scholars proposed the domain adaptive object detection (DAOD) methods, these methods mostly rely on convolution as the basic operation, which is local and spatially invariant and cannot model the overall spatial variations in the images. When there are changes in the overall image characteristics, wrong detections in DAOD are illustrated in Fig. 3. The red boxes indicate instances of missed detections. As shown in Fig. 3(a), the impact of raindrops leads to the inability to detect cars. In Fig. 3(b), the model's detection accuracy decreases in another underwater environment due to differences in lighting conditions and color changes caused by large bodies of water (such as the ocean), resulting in the inability to detect echinus.

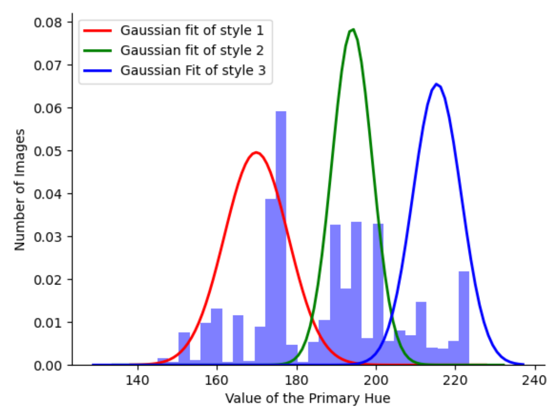


Fig. 1. The distribution of hue values in the Deepfish dataset.

As it is well known, image representations in the frequency domain exhibit fixed patterns [6], where high frequency components represent image textures and details, and low-frequency components represent flat and smooth areas [7]–[11]. Each frequency component is computed based on a global receptive field, achieving performance similar to that

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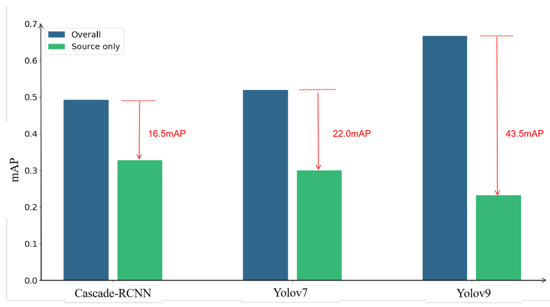



Fig. 2. Comparison of detection accuracy.

of multi-head attention [12]. The lower computational cost makes it to be a better choice, especially in scenarios with limited computational resources. Inspired by these, an efficient frequency-based attention mechanism is developed. We design adaptive frequency filters to identify Domain-Invariant Frequencies (DIF) and Domain-Varying Frequencies (DVF) capturing feature representations that contribute to model's generalization performance.

To train learnable frequency filters, we adopt the strategy of adversarial feature learning and developed MIFA module. This module consists of multiple domain classifiers connected to different scale convolution blocks. A gradient reversal layer (GRL) [13] is applied during training to reverse the gradient direction, which helps in distinguishing frequency information to retain, facilitating potential feature selection. Simultaneously, it encourages the backbone network to generate domain-confused features, reducing the distribution gap between the target domain and the source domain. Experimental results show that the proposed method achieves high accuracy and efficiency.

The main contributions of our paper can be summarized as follows:

- We introduce a object detection model FIOD-VUE that focuses on domain-invariant information in changing underwater environments, which can be applied to any second-order object detector. The model has effective generalization ability in UOD tasks.
- We develop an efficient domain-invariant information attention mechanism FIA based on the frequency domain. By employing learnable filters, our approach achieves adaptive identification of DIF and DVF components in the frequency domain. Integrating frequency-aware cues into CNN models facilitates modeling of global information, enabling adaptation to variable environment across images.
- We propose MIFA based on adversarial feature learning strategy, which is used only during training. MIFA facilitates a minimax game between the multi-scale domain classifier and the detection network, training both the frequency filter and the backbone network. By minimizing the difference between the distributions of source and target domain data, MIFA enhances the model's generalization capability without compromising its efficiency.
- We construct the HD-Deepfish dataset by reorganizing the Deepfish dataset using hue clustering, which provides an additional validation dataset for underwater adaptive object detection.
- Extensive experiments across domain adaptation tasks in different environments validate the effectiveness and superiority of our proposed approach.

The remainder of the paper is organized as follows. Section II reviews and summarizes the current methods. Section III introduces the framework of the proposed FIOD-VUE network. Sections IV and V conduct the experiments, comparative analysis against state-of-the-art models, and discuss the results. Finally, we concludes this work in Section VI.

## II. Related Work

In this section, we provide a brief review of prior research from three perspectives: domain adaptive object detection, underwater object detection, and frequency domain learning.

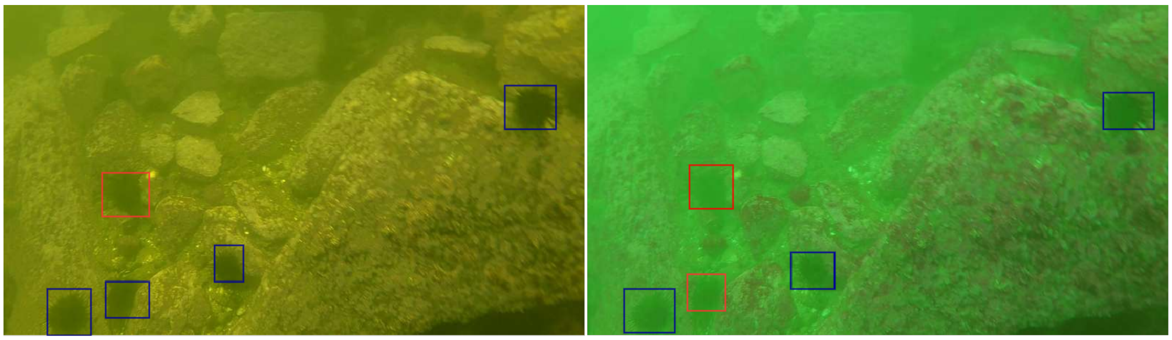
### A. Domain Adaptive Object Detection

DAOD has been extensively researched in recent years with the primary aim of reducing the domain discrepancy between the source and target domains, minimizing domain classification errors. Most existing domain adaptive detectors employed CNN-based detectors and implement strategies such as adversarial feature learning [14]–[19], self-training with pseudo labels [20]–[22], domain randomization [23]–[25], and mean teacher training [26], [27]. Adversarial feature learning uses GRL to reverse gradients during training. This process generates deceptive features for domain classifiers, resulting in domain-invariant features for detectors. For instance, domain adaptive Faster R-CNN (DAF) [28], a pioneering work in DAOD, combined Faster R-CNN with adversarial training to adjust distributions of image and instance level cross domains. A series of improvements have been made on this basis, such as conducting gradient alignment in adversarial training [29], adding prior adversarial losses [15], enhancing local alignment [30], and increasing object scale during training [31]. Some methods utilized global frequency features to assist the domain adaptation process. For example, the multi-view adversarial discriminator (MAD) [32] divided images into causal and non-causal parts through fixed bandpass filtering. It then randomized non-causal parts to increase source domain diversity. Different from MAD, we introduce spatial attention in the frequency domain, which is a learnable filters. Through adversarial training, it enable weighted processing of different frequency components. Our method shifts the focus of the model towards domain-invariant features, thereby achieving effective domain adaptation.

### B. Underwater Object Detection

Due to the complex and variable nature of underwater environments, collected underwater images typically exhibit different color biases, leading to a sharp decrease in detection performance. Existing methods for addressing this issue in UOD can be broadly categorized into data augmentation methods [23], [33]–[36], prior information assistance [37], and domain adaptation methods [38]. Many researchers adopted generative adversarial models for style transfer, thereby achieving

(a) Street Scene. Green boxes indicate correctly detected cars, while red boxes indicate missed detections.



(b) Underwater Scene. Blue boxes indicate correctly detected echinus, while red boxes indicate missed detections.

Fig. 3. Visualization examples of poor detection results.

data augmentation. For instance, Li et al. [33] designed a generative adversarial network to generate realistic underwater images from aerial images and depth maps. Fabbri et al. [34] directly employed CycleGAN to generate underwater images. Li et al. [35] synthesized ten different underwater images with varying scene parameters. Chen et al. [23] generated images of different styles by conditional bilateral style transfer and achieved cross-domain generalization through contrastive learning focusing on domain-invariant features. However, models trained on synthetic data often struggle to generalize well to real underwater scenarios. Song et al. [37] used prior probability reasoning to help improve the applicability in underwater scenes. Some researchers considered using domain adaptation methods such as adversarial structures to enhance the adaptive capability of UOD models in complex underwater environments. For example, Liu et al. [38] proposed adversarial feature learning and introduced the domain-generalized UOD task. This method enhanced the model's feature extraction capability. However, it was still in the early stages of addressing the challenges posed of complex underwater environments. It had yet to fully capture the key characteristics of different underwater images.

### C. Frequency Domain Learning

Frequency analysis has been proven to be an effective tool in the field of signal processing [39]. In spectral data, each frequency component is correlated with other frequency components, hence each frequency has a global receptive field. Low-frequency components mainly capture domain-specific information such as color and illumination [40], while high-frequency components primarily represent domain-invariant information, such as edges and shapes, which are characteristics of objects [41]. In object detection tasks, the contributions of high-frequency and low-frequency components differ to the model's generalization and accuracy. Qin et al. [42] proposed a frequency channel attention network (FcaNet), mathematically proving that traditional global average pooling is a special case of frequency domain feature decomposition. Zhong et al. [43] achieved performance beyond human biological vision by mining object cues in the frequency domain with the help of discrete cosine transform. Zhou et al. [44] improved the feature representation of images from a frequency perspective, providing stronger frequency cues for the boundary-aided task. These relevant works sparkled the inspiration of modulating different frequency components to improve the generalization of the UOD across different environments.

## III. Methodology

### A. Overview

A detailed overview of the overall framework is illustrated in Fig. 4. We build our approach on the basis of Cascade RCNN [45]. Firstly, the input image is processed through the FIA module as illustrated in the left part of Fig. 4. The details of the FIA module are discussed in Section B. The input image and the processed image are then fused through residual connections. Subsequently, ResNet serves as the feature extraction backbone to extract four scales of feature maps, ranging from shallow to deep levels. MIFA operates on these feature maps, where domain adversarial training is realized through GRL. This layer automatically reverses gradients during propagation to achieve domain-confusion features with reduced domain differences in the backbone network, as depicted in the bottom right part of Fig. 4. The details of MIFA are elaborated in Section C. Finally, we devise an object detection loss

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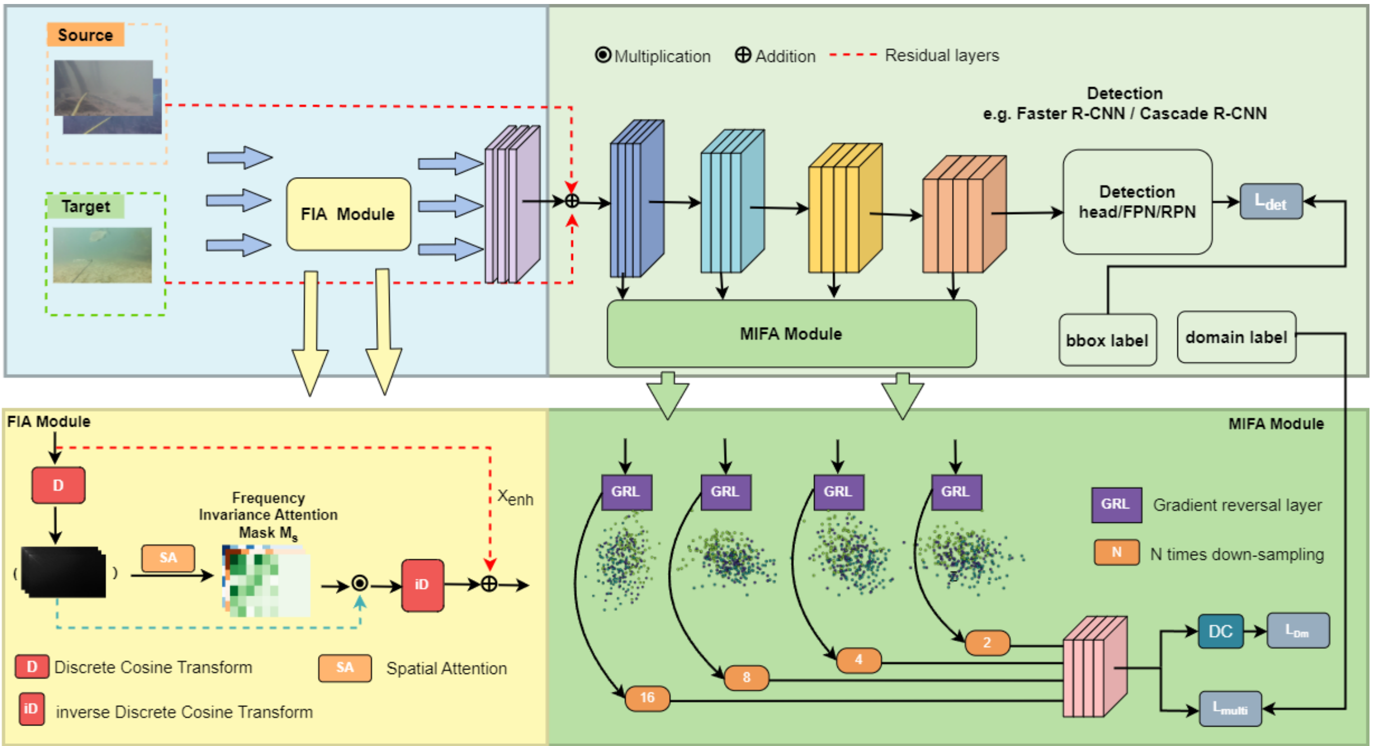


Fig. 4. The overall structure of FIOD-VUE can be divided into three components: (1) Yellow Part: Adopts Frequency Invariance Attention (FIA) to adaptively focus on DIF. (2) Light Green Part: Utilizes the object detection model to extract multi-level features, predicting object positions and categories. (3) Dark Green Part: Executes Multi-scale Image-level Feature Alignment (MIFA), engaging in a game between the domain classifier and the backbone network to optimize the backbone network.

that focuses on invariant information in the ever-changing underwater environment through the collaboration of object detection loss, adaptation loss, and multi-scale alignment loss, as discussed in Section D. It is worth noting that the proposed method can be applied to any second-order object detection network.

### *B. Frequency Invariance Attention (FIA)*

In FIA, the frequency components of the input image are filtered through a frequency-based attention mechanism to enhance domain-invariant features and suppress domain-changing features. Given an image signal $X(x, y)$ in the spatial domain, with an image height $H$ and width $W$. The basis functions for two-dimensional Discrete Cosine Transform (2D DCT) are defined as follows [46]:

$$B_{u,v}^{x,y} = cos\left[\frac{\pi\,(2x+1)\,u}{2H}\right] cos\left[\frac{\pi\,(2y+1)\,v}{2W}\right] \tag{1}$$

for $u, x = 0, 1, \ldots, H-1$, and $v, y = 0, 1, \ldots, W-1$. Then, the 2D DCT and its corresponding inverse transform can be expressed as follows:

$$X_f(u,v) = \frac{2}{\sqrt{HW}} \sum_{x=0}^{H-1} \sum_{y=0}^{W-1} C(u)C(v)g(x,y)B_{u,v}^{x,y} \tag{2}$$

$$X(x,y) = \frac{2}{\sqrt{HW}} \sum_{u=0}^{H-1} \sum_{v=0}^{W-1} C(u)C(v)G(u,v)B_{u,v}^{x,y} \tag{3}$$

In both equations (2) and (3) $C(u)$ is defined as:

$$C(u) = \begin{cases} \frac{1}{\sqrt{2}} & u = 0 \\ 1 & u \neq 0 \end{cases}, C(v) = \begin{cases} \frac{1}{\sqrt{2}} & v = 0 \\ 1 & v \neq 0 \end{cases} \tag{4}$$

For each image $X^i \in \mathbb{R}^{H \times W \times 3}$, an independent DCT is applied to each channel, regardless of whether it comes from the source domain or the target domain, where $i$ represents the position of the image in the batch. This transformation can be expressed as $X_f^i = DCT(X^i)$. The frequency representation $X_f^i$ can be transformed back to the original spatial domain using the inverse Discrete Cosine Transform (iDCT), succinctly expressed as $X^i = iDCT(X_f^i)$. For $X_f^i$, the different frequency components of the original image information are decomposed into elements at different spatial positions within $X_f^i$. This can be viewed as a frequency-based decoupling and rearrangement of $X^i$, enabling us to achieve frequency filtering through spatial attention modules. Moreover, $X_f^i$ serves as a natural global feature representation that can facilitate the suppression of specific domain information distributed globally, such as changes in lighting and color cast tones. This, in turn, enhances the object information, specifically the domain-invariant information. Therefore, a frequency attention mechanism for learning domain-invariant information is proposed, which can be described as follows:

$$X_f^{i\,\prime} = X_f^i \otimes M_S(X_f^i) \tag{5}$$

Where $\otimes$ denotes element-wise multiplication. $M_S(\cdot)$ is the spatial mask learned by the spatial attention module. The

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global max pooling and global average pooling are applied along the channel dimension of input image $X_f^i$ and two feature maps of size $H \times W \times 1$ are obtained. Concatenate the results of global max pooling and global average pooling along the channel dimension to obtain a feature map of size $H \times W \times 2$. Finally, apply a $7 \times 7$ convolution operation to the concatenated result to obtain a feature map of size $H \times W \times 1$, and then apply a Sigmoid activation function to obtain a spatial mask $M_S(\cdot)$ of size $H \times W \times 3$. The process is represented as follows:

$$M_S(X_f^i) = \sigma(Conv([AvgPool(X_f^i); MaxPool(X_f^i)])) \quad (6)$$

Here, AvgPool and MaxPool represent the average-pooled and max-pooled features over the entire channel, $[\cdot;\cdot]$ denotes an aggregation operation, Conv signifies convolution with a filter size of $7\times7$, and $\sigma$ represents the sigmoid function. Finally, we restore the frequency-space modulated features back to the original spatial domain through the inverse Discrete Cosine Transform (iDCT) to obtain $X_{enh}$. This transformation can be expressed as $X_{enh}^i = X^i + iDCT(X_f^{i\prime})$, $X_{enh}^i \in \mathbb{R}^{H \times W \times 3}$.

### C. Multi-scale Image-level Feature Alignment (MIFA)

Considering that the domain shift between different underwater environments is primarily attributed to background color deviation, the MIFA is compounded to calibrate the distribution differences between domains in the convolutional feature maps. An implicit assumption is that if the distribution between domains in deep feature maps is similar, then the distribution between domains in shallow feature maps is also roughly the same [47]. To minimize domain discrepancies between different scales, we design the MIFA module with adversarial domain classifier sub-modules within different convolutional blocks. The adversarial domain classifier confuses domain features, and optimization is performed through a minimax game between the domain classifier and the backbone network.

MIFA consists of four domain classifiers, each comprising three $1 \times 1$ convolutional layers followed by a softmax function. These classifiers categorize the feature maps from the four stages of the ResNet. Between each domain classifier and the backbone network, there is a GRL. During forward propagation, the GRL acts as an identity transformation. However, during backpropagation, GRL takes the gradient from the subsequent layer and multiplies it by -1 before passing it to the previous layer. For $X_{enh}^i$, the domain features of the m-th block's convolutional layer can be represented as $F_m^i, m \in \{1, 2, 3, 4\}$. The adversarial classifier submodule for the m-th block is denoted as $D_m$, and it is used to predict the domain label of $F_m^i$. The optimization objective for domain classification can be expressed as:

$$\max_{\mathcal{W}} \min_{D_m \in \mathcal{H}} L_{\mathcal{D}_m} \quad (7)$$

where $\mathcal{H}$ represents the hypothesis space of the domain classifier, $\mathcal{W}$ is the feature extractor network ResNet, and $L_{\mathcal{D}_m}$ denotes the domain classification error in the m-th convolutional block as follows:

$$L_{\mathcal{D}_m} = -\frac{1}{N_d X_h X_w} \sum_{k,x,y} l_k^i D_m \left( F\left(X_{enh}^i\right)^{(x,y)} \right) \quad (8)$$

where i signifies the i-th image within the batch, and the variables x, y, and k are defined as follows: $x \in \{0, \ldots, X_h - 1\}$, $y \in \{0, \ldots, X_w - 1\}$, and $k \in \{0, \ldots, N_d - 1\}$. $l_k^i$ represents the domain label corresponding to the k-th domain classifier obtained from the i-th image. $X_h$ and $X_w$ represent the height and width of the feature map, respectively, and $N_d$ denotes the number of domains. $F(X_{enh}^i)^{(x,y)}$ indicates a 1 × C feature at the position (x, y) within the feature map.

To minimize domain discrepancies across different scales, we further apply a consistency regularizer, as expressed below:

$$L_{multi} = \frac{1}{|I|\, C_M^2} \left\| \sum_{m,n} \sum_{x,y} \left( p_m^{(x,y)} - p_n^{(x,y)} \right) \right\|_2 \quad (9)$$

The outputs of the four domain classifiers are downsampled separately, with downsampling rates of 16, 8, 4, and 2, respectively. Where $p_m^{(x,y)}$ and $p_n^{(x,y)}$ represent any two different downsampling results. $\|\cdot\|$ denotes the $L_2$ distance, $|I|$ represents the total number of activations in the feature map, and $C_M^2$ is the sum of all pairs of convolutional blocks.

### D. Total Objective Loss Function

The overall objective loss function is the summation of detection loss and adaptation loss:

$$L = L_{det} + \lambda_1 L_{D_m} + \lambda_2 L_{multi} \quad (10)$$

$\lambda_1, \lambda_2$ are weighting factors that balance the detection loss, adaptation loss, and multi-scale alignment loss. We denote the detection loss as $L_{det}$, which encompasses the losses from the region proposal network along with the final classification and localization errors. $L_{D_m}$ and $L_{multi}$ correspond to the adversarial loss and consistency regularization term discussed in Section C.

## IV. Datasets and Experiments

For underwater environment, the absorption and attenuation of light by water cause color bias and consequently impacting object detection and recognition. In order to better study this type of problem, we construct a new domain adaptation object detection dataset that spans multiple underwater environments, named HD-Deepfish. Then we compare our approach with other DAOD methods such as DAF [14], SW-DA [15], GPA [18], Center-Aware [19], MAD [32] as well as UOD methods like DG-YOLO [38], RoIAttn [36], DMCL [23], and Boosting R-CNN [37] to demonstrate its superior performance. Following that, we conduct comprehensive ablation experiments to validate the effectiveness of each module proposed in FIOD-VUE. Lastly, we provide an analysis of feature statistics and visualizations, which indicate that our method contributes to learning domain-invariant features.

### A. Datasets

There are few publicly available UOD datasets that categorize different domains based on the changing underwater environments. Chen et al. proposed S-UODAC2020 [23] which is a domain generalization dataset for the UOD. However, in the S-UODAC2020 dataset, all domain types are manually selected, and the division is arbitrary, leading to strong similarities between some domains. To construct available datasets for validating UOD generation across domains, we propose the HD-Deepfish dataset based on adaptive hue biases.

Firstly, we convert each underwater image in Deepfish dataset [48] to the HSV space and identify the dominant hue, denoted as $H_i, i$, as the hue with the highest pixel proportion in the image. We compute the $H_i$ values for all underwater images and plotted the histogram, as shown in Fig. 1. Then, we apply the k-means algorithm to cluster the results. The clustering metrics are presented in Table I. As listed in Table I, for the clustering number k=3, the Silhouette Score (SS) [49] is closest to 1, the Calinski-Harabasz index (CH) [50] is maximal, and the Davies-Bouldin index (DB) [51] is minimal. The resulting k=3 clusters are referred to as style1, style2, and style3 in the subsequent discussion.

TABLE I
CLUSTERING RESULTS OF THE DOMAIN COLOR TONES IN HD-DEEPFISH.

| K | SS | CH | DB |
|---|---|---|---|
| 3 | 0.69 | 31253.75 | 0.40 |
| 4 | 0.66 | 20883.49 | 0.46 |
| 5 | 0.68 | 30466.81 | 0.44 |

To maintain a balance among the three styles, we apply data augmentation techniques such as flipping, rotation, and jittering to enhance the style1 dataset. Totally, there are 2029 images in style1, 2546 images in style2, and 2585 images in style3, totaling 7160 images. Fig. 5 shows the samples of the HD-Deepfish Dataset. Our dataset is available at: https://github.com/JOU-UIP/HD-Deepfish.

In addition to S-UODAC2020 and HD-Deepfish, We also utiliz the Cityscapes dataset series as supplementary experiments to validate the generalization capability of FIOD-VUE. The Cityscapes dataset series comprises Cityscapes [52], Foggy Cityscapes [52], and Rain Cityscapes [53]. Foggy Cityscapes and Rain Cityscapes are derived from the Cityscapes dataset by simulating fog and rain on the clear weather images from Cityscapes. Both urban and foggy cityscapes have the same number of categories, including person, rider, car, truck, bus, motorcycle, and bicycle.

### B. Experimental Setup

Consistent with domain adaptation for object detection, the source domain's annotated objects are used to train the detection module, while the target domain's objects have no annotations whatsoever. Unlike typical domain adaptation object detection models, our source domain has more than one domain label. We split our data into training and testing sets in a ratio of 7:3 to train and evaluate our method. The validation set is separated from the training set, with a ratio of 10%. In the comparative experiments, all models are tested on the same target domain test set. Our FIOD-VUE is implemented in both PyTorch and the MMDetection framework, with all experiments conducted on an NVIDIA GeForce RTX 4090 GPU, with model training taking approximately 2.5 hours. Throughout all experiments, input images are first resized to a shorter side length of 800 before being fed into the network model. We adopt ResNet50 pretrained on ImageNet as the backbone, and train it using SGD optimizer with an initial learning rate of 0.001. A total of 24 epochs are trained, with the learning rate reduced by a factor of 10 at the 22nd and 23rd epochs. In the supplementary materials, we provide the loss curve plots to demonstrate the convergence of the model during training. The batch size is set to 2, with images randomly sampled from both the source and target domains. The balancing parameters, $\lambda_1$ and $\lambda_2$, are set to 0.1. We report mean average precisions (mAP) with an IoU threshold of 0.5.

### C. Experiments Across Underwater Environments

The results on S-UODAC2020 are shown in Table II. Our method achieved the best object detection results, with an mAP of 56.8% on validation set type 7, as shown in Fig. 5. This indicates that the model effectively filters out components that hinder cross-domain adaptation and better narrows the gap between the source and target domain distributions. The learned domain-invariant information plays a crucial role in the cross-domain detection performance. Our method outperforms UOD methods like RoIAttn and BoostingR-CNN, which focus only on detecting objects consistent with the training set distribution in the target domain. Compared to UOD methods used for domain generalization, such as DG-YOLO and DMCL, our method performs better, demonstrating that the FIA method can further adaptively enhance domain-invariant information to accommodate different underwater conditions. In practice, underwater exploration often faces limitations in platform resources. Therefore, we compared the model complexity (FLOPs), detection speed (FPS), and model parameters (Params) of different methods. Boosting R-CNN has the lowest complexity and the least number of parameters, but its mAP is only 24.8% and 27.1%. Considering these three metrics, DMCL performs the best. Our proposed method has a 5% lower FPS and a 17% reduction in FLOPs compared to DMCL, with a 15% increase in parameters. Overall, the performance is roughly comparable.

We conduct three-fold cross-validation experiments when testing the model's generalization ability on the constructed HD-Deepfish dataset, and the results are presented in Table III. It can be observed that our method outperforms traditional UOD methods in terms of generalization ability on datasets with variations in underwater color cast due to different lighting and water quality conditions. In comparison with DG-YOLO and DMCL, our approach focuses more on image-level domain generalization, specifically color cast deviation, which is more suitable for instance-level domain generalization methods. It's worth noting that the detection results of most models are relatively poor in style2. We believe this could be due to the blue color bias, which makes it more challenging to distinguish between the object and the surrounding environment, thereby affecting the detection results.

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Fig. 5. HD-Deepfish Dataset. The left column corresponds to style1, the middle column to style2, and the right column to style3.

TABLE II
PERFORMANCE OF DIFFERENT UOD MODELS ON THE S-UODAC2020 DATASET. TRAINED ON STYLE1-STYLE6 AND TESTED ON STYLE7. AP50 REPRESENTS IOU=0.5, AP75 REPRESENTS IOU=0.75, MAP REPRESENTS IOU=0.5:0.05:0.95.

| Method | Echinus | Starfish | Holothurian | Scallop | AP50 | AP75 | mAP | FLOPs | FPS | Params |
|---|---|---|---|---|---|---|---|---|---|---|
| DG-YOLO [38] | 70.3 | **62.5** | 44.6 | 43.5 | 62.7 | 56.5 | 50.4 | 262.5 | 4.2 | 61.5 |
| RoIAttn 50 [36] | 33.2 | 20.9 | 20.9 | 20.2 | 50.5 | 18.5 | 23.8 | 445.6 | 11.2 | 55.2 |
| RoIAttn 101 [36] | 32.0 | 18.8 | 19.4 | 21.2 | 48.6 | 18.0 | 22.8 | 521.6 | 9.3 | 74.2 |
| DMCL [23] | **80.1** | 35.5 | 47.7 | 43.5 | 65.9 | 58.4 | 56.5 | 282.8 | **17.7** | 60.1 |
| Boosting R-CNN 50 [37] | 33.8 | 21.4 | 20.8 | 23.2 | 51.9 | 20.4 | 24.8 | **173.2** | 17.0 | **45.9** |
| Boosting R-CNN 101 [37] | 36.0 | 20.0 | 25.6 | 26.7 | 53.2 | 23.8 | 27.1 | 235.0 | 14.1 | 68.3 |
| Ours | 49.4 | 61.5 | **54.8** | **61.6** | **91.6** | **91.6** | **56.8** | 234.6 | 16.7 | 69.5 |

TABLE III
PERFORMANCE OF DIFFERENT UOD MODELS ON THE HD-DEEPFISH DATASET.

| Method | style 1,2 | style 1,3 | style 2,3 |
|---|---|---|---|
| DG-YOLO [38] | 27.8 | 22.8 | 18.9 |
| RoIAttn 50 [36] | 30.1 | 33.2 | 32.3 |
| RoIAttn 101 [36] | 36.3 | 20.3 | 24.3 |
| DMCL [23] | 34.7 | 29.4 | 28.4 |
| Boosting R-CNN 50 [37] | 34.8 | 20.6 | 18.3 |
| Boosting R-CNN 101 [37] | 36.8 | 25.6 | 34.6 |
| Ours | **37.1** | **35.8** | **36.2** |

### D. Experiments Across Other Meteorological Environments

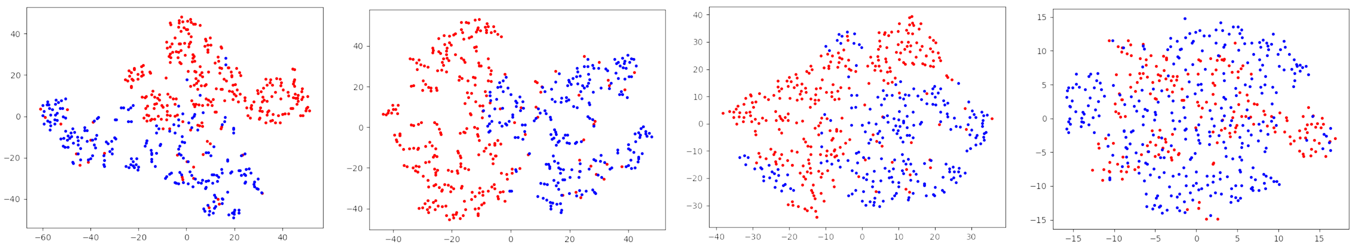
We also compare our FIOD-VUE with mainstream DA methods on Cityscapes dataset series to validate the model's generalization ability under different weather conditions. We train on the Cityscapes and Foggy Cityscapes datasets and test on the Rain Cityscapes dataset. The results are presented in Table IV. The results indicate that our method exhibits better generalization capabilities under specific weather conditions. In particular, our approach outperforms all other methods in the truck category, where frequency-aware cues aid in distinguishing between cars and trucks. It underscores that the effectiveness of our method not only in environments with underwater color cast variations but also in other challenging weather conditions.

### E. Visualization of t-SNE

We use t-SNE projection to visualize the similarity distributions of features. We randomly select 300 images in HD-Deepfish from the source domain and 100 images from the target domain and project the features from the final stage of the FIOD-VUE's backbone using different methods onto a 2D plane, as shown in Fig. 6. Blue points represent data from the source domain, while red points represent data from the target domain. If the red and blue points are well integrated,

TABLE IV
TESTING RESULTS ON THE CITYSCAPES DATASET SERIES.

| Method | person | rider | car | truck | bus | motorcycle | bicycle | mAP |
|---|---|---|---|---|---|---|---|---|
| DAF [14] | 19.5 | 34.9 | 41.8 | 9.6 | 58.7 | 7.6 | 18.3 | 27.2 |
| SW-DA [15] | 25.0 | 60.4 | 51.6 | 5.0 | 67.3 | **24.2** | 32.2 | 33.2 |
| GPA [18] | 26.6 | 60.4 | 55.3 | 6.3 | 67.5 | 14.8 | **37.9** | **38.4** |
| Center-Aware [19] | 28.2 | **65.6** | 45.5 | 11.7 | 65.6 | 15.8 | 31.7 | 37.7 |
| MAD [32] | **28.5** | 61.4 | **60.8** | 15.7 | 70.4 | 10.6 | 13.5 | 37.3 |
| Ours | 26.6 | 44.7 | 55.0 | **34.0** | **71.6** | 13.0 | 24.2 | **38.4** |



(a) DG-YOLO (b) RoIAttn 50 (c) Boosting R-CNN 50 (d) Ours

Fig. 6. t-SNE visualizations for different methods

TABLE V
ABLATION RESULTS OF EACH COMPONENT TRAINED AND TESTED ON HD-DEEPFISH DATASET.

| SA | FIA | IFA | MIFA | PS | $L_{D_M}$ | $L_{multi}$ | mAP | FLOPs | FPS | Params |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | 27.6 | 234.51 | 18.9 | 68.97 |
| ✓ | | | | | | | 29.6 | 234.56 | 18.6 | 68.97 |
| | ✓ | | | | | | 31.3 | 234.61 | 16.7 | 68.97 |
| | ✓ | ✓ | | | ✓ | | 32.8 | 234.61 | 16.7 | 69.08 |
| | ✓ | | ✓ | ✓ | ✓ | | 33.4 | 234.61 | 16.7 | 69.08 |
| | ✓ | | ✓ | | ✓ | | 34.6 | 234.61 | 16.7 | 69.48 |
| | ✓ | | ✓ | | ✓ | ✓ | **37.1** | 234.61 | 16.7 | 69.48 |

it indicates that the model captures more domain-invariant features. The results in Fig. 6 indicate the effectiveness of our method compared to other models in terms of feature alignment.

In addition, we visualize the features obtained by each module to analyze the role of each module in feature alignment, as shown in the Fig. 7. The results show that the blue and red dots in the input image are mostly non-coalescent, as shown in Fig. 7 (a). In the frequency domain, the representation of the image has fixed patterns. After passing through the frequency filters of IFA, the interesting frequency components are captured, and there is a tendency for the blue and red dots to merge. However, it can be seen that the data still maintains its original clustering structure in the feature space, as shown in Fig. 7 (b). After adversarial training with MIFA, the feature extraction modules at multiple scales obtain a certain degree of semantic consistency. Most of the red and blue points are further blended and overlapped. It is noteworthy that the structures of the two clusters also begin to approach each other as much as possible. The results of Fig. 7 (c) demonstrate that our method effectively alleviates the effects of domain shift.

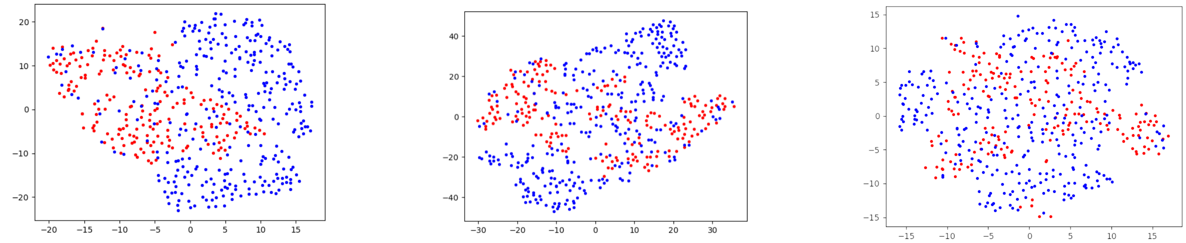
### *F. Ablation Study*

We perform ablation experiments on the HD-Deepfish dataset, testing different configurations. The findings are consolidated in Table V. Our aim is to discern the impact of each component by gradually integrating them and noting shifts in mAP performance alongside complexity. We juxtapose the direct utilization of spatial attention (SA) with FIA to gauge FIA's efficacy in enhancing detection performance. IFA refers to connecting a domain classifier after the last convolutional block, while MIFA refers to connecting a domain classifier after each convolutional block. Parameter sharing (PS) indicates whether the parameters of the four domain classifiers are shared. Comparisons between single-scale and multi-scale domain classifiers suggest that minimizing domain differences between image-level features at different scales helps further enhance the model's semantic alignment capability. By further applying consistency regularization, our model achieve an mAP of 37.1%. Since domain classifiers are only used during training, the overall computational increment only occurs in the FIA part, with an increase of only 0.1M operations, while achieving a 9.5% increase in mAP.

We also tested the balance coefficients $\lambda_1$ and $\lambda_2$ for the domain adversarial loss, ranging from 0.05 to 0.2 with intervals of 0.05. As shown in Fig. 8, we set $\lambda_1 = 0.1$ and $\lambda_2 = 0.1$ for all experiments in our FIOD-VUE model.

## V. CONCLUSIONS

This paper focuses on the task of UOD, particularly addressing the domain shift problem in underwater environments with changing color casts in real-world scenarios. This remains a meaningful yet underexplored challenge. We introduce the



(a) Features of Input (b) Features of IFA (c) Features of MIFA

Fig. 7. t-SNE visualizations for different modules

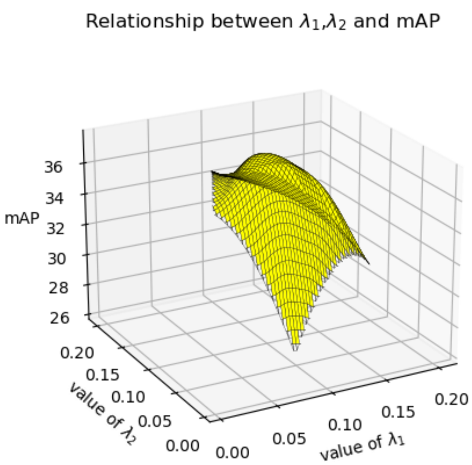



Fig. 8. Results of hyperparameter ablation for balancing factors $\lambda_1$ and $\lambda_2$.

HD-Deepfish dataset which is an additional validation dataset for underwater adaptive object detection. Furthermore, we propose the FIOD-VUE model, which includes a FIA component that represents the image-level features of color distortion from a global perspective, autonomously distinguishing DIF and DVF in the frequency domain. This models the image-level characteristics of color distortion from a global perspective. Additionally, we introduce the MIFA component, which enables adaptive learning of frequency filters through adversarial methods. Experimental results demonstrate that FIOD-VUE showcase effectiveness in varying underwater enviroment and other challenging weather conditions. We believe our method can advance research on the generalization task in UOD, enhancing its adaptability in practical applications.

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